

Yasha Duncan-Wilson

yasha@duncan-wilson.co.uk | github.com/mechanical-girl

Education

Heriot-Watt University – MEng in Aerospace Engineering

2028

Experience

Team Lead Heriot-Watt University CanSat Society – Edinburgh

2024 – Present

Mechanical Engineer Heriot-Watt University CanSat Society – Edinburgh

2023 – 2024

- Selected and lead a team to compete in the AAS CanSat Competition, managing budget, deadlines and team dynamics as well as engineering requirements.
- Performed analyses of competing solutions to complex problems, ultimately choosing designs which were fit for purpose while meeting mass and material restrictions.
- Adapted aerospace industry design patterns and analytic tools such as FMEA and RCA for use by a small team.

Structures Engineer Team AerooWatt – Edinburgh

2024 – Present

- Designed and fabricated new light-weight wing structures for autonomous fixed-wing aircraft.
- Proposed novel methods for mass reduction and launch reliability.

Summer Intern Heriot-Watt University GRID facility – Edinburgh

2024

- Designed and implemented solution for reliable automatic 3D printer bed clearing with no intervention.
- Improved throughput by reducing the need for lab technicians to clear prints and reset printers.

Projects

Ender 3 Klipper Conversion

<https://github.com/mechanical-girl/klipper-ender-3>

- Improved poor print success rates: built and fitted an ABL probe to improve first-layer reliability
- Upgraded firmware: wrote and tuned Klipper config to improve print outcomes and user experience

Skills

Programming: Python

3D Printer Firmware: Marlin, Klipper

CAD: Solidworks, Fusion 360